

No Model Wins Everything: A Multi-Objective Evaluation of a Nested Hybrid Movie Recommender

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Abstract

Recommender systems are usually reported with a single headline number, and the choice of that number decides which system looks best. This project builds seven movie recommenders on MovieLens ml-latest-small and evaluates all of them on one held-out split under five families of measure: rating error, ranking quality, catalogue coverage, beyond-accuracy quality (novelty and serendipity), and cold-start reach. The models are a damped-bias SVD, an item-based autoencoder (I-AutoRec), a TF-IDF content model over TMDB plot overviews, three hybrid compositions of those, and a non-personalized Most-Popular baseline. The central result is that no model wins everything: across six measures, three different models hold the six titles. I-AutoRec attains the lowest rating error (RMSE 0.8517) while recommending from 0.6% of the catalogue, a figure identical to the non-personalized baseline's, and its novelty of 1.86 sits nearer that baseline's 1.65 than the 7.13 a uniformly random recommender would score. The content model is the most diverse (24.9% coverage, novelty 6.76, 481 cold-start slots) and the worst ranker (NDCG@10 0.0359). A nested hybrid, a recommender whose first component is itself a hybrid, ranks best (NDCG@10 0.1643). Because a single split cannot separate small differences, the entire comparison is repeated across four splits: eight of ten reported claims hold on every split, while the claim that SVD fails to beat the popularity baseline does not. That paired difference is +0.0012 NDCG@10 and changes sign twice, so collaborative filtering alone is better described as indistinguishable from recommending best-sellers than as worse than it. The weighted hybrids, by contrast, beat plain SVD on every split by $+0.0103 \pm 0.0022$.

1. Introduction

A catalogue of ten thousand films is larger than any viewer can inspect, so the practical question is not whether to filter it but how. Recommender systems answer that by estimating preference from past interactions and ranking the catalogue accordingly. The research literature has converged on a small set of measures for judging how well they do it — rating error for predicted scores, and precision, recall and normalized discounted cumulative gain for ranked lists — and a system is normally reported by one of them.

That convention hides something. A model can score well on rating error by predicting confidently on the films almost everyone has already rated, and a ranking metric can only credit a recommendation whose item somebody in the test set happened to rate. Both measures therefore reward concentration on popular items, and neither penalises a system that never recommends anything else. Whether that matters is an empirical question, and it is the question this project asks.

Research question. *Can a hybrid recommender improve on its individual components, and does “improve” mean the same thing when quality is measured by accuracy, by catalogue behaviour, and by novelty?*

Four objectives follow from it:

1. Implement and evaluate three recommenders of different families (matrix factorization, deep autoencoder, content-based NLP) on a common split.
2. Compose them into hybrids by weighted blending, switching and nesting, rather than treating a three-model blend as a new algorithm.
3. Measure all seven systems on the same held-out data under rating error, ranking, coverage, novelty, serendipity and cold-start reach.
4. Quantify how far the resulting conclusions depend on the particular train/test split they were measured on.

The contributions are:

- A composable hybrid in which a hybrid is itself a valid component, so a three-model system is a two-line composition rather than a rewrite (Section 4.4).
- A seven-model comparison under five metric families, showing that three different models hold the six available titles (Section 5.5).
- Evidence that the deep model’s advantage in rating error is bought by collapsing onto popular items: its catalogue coverage equals the non-personalized baseline’s to one decimal place, and its novelty is closer to that baseline than to chance.
- A proof that cold-start failure here is a property of the data rather than of model capacity: for items with no training ratings, both the linear and the non-linear model produce predictions with exactly zero variance (Section 5.6).
- A robustness check that repeats the whole comparison on four splits and identifies which reported claims survive (Section 5.8).

2. Related Work

2.1. Collaborative filtering and matrix factorization

Collaborative filtering estimates preference from the user–item interaction matrix alone. Sarwar et al. (2001) moved the neighbourhood computation from users to items, arguing that item–item relationships are more stable and can be precomputed. Matrix factorization replaced explicit neighbourhoods with latent factors; Koren et al. (2009) set out the formulation used here, including the user and item bias terms that account for the systematic tendency of some raters to rate high and some films to be rated high. This project uses a truncated singular value decomposition of the mean-centred matrix with a damped item-bias term, which is the same idea in its simplest form.

2.2. Deep collaborative filtering

Autoencoder and neural formulations aim to capture interaction patterns that an inner product cannot. Sedhain et al. (2015) proposed AutoRec, an autoencoder that reconstructs a partially observed rating vector under a loss masked to the observed entries; the item-based variant, I-AutoRec, treats each item’s vector of user ratings as one training example and is the variant implemented here. He et al. (2017) generalised the idea, replacing the inner product with a learned interaction function. Both report accuracy gains over matrix factorization. Neither is evaluated on catalogue behaviour, which is the gap this project addresses for I-AutoRec specifically.

2.3. Content-based and hybrid recommendation

Content-based methods score items by similarity to what the user has already liked, using item descriptions rather than co-rating patterns, and so remain defined for items with no interaction history at all. Burke (2002) provides the taxonomy this project follows directly: the weighted hybrid combines component scores numerically, while the switching hybrid selects a component per case according to a criterion. The nesting used in Section 4.4 is a composition of weighted hybrids rather than a new category in that taxonomy.

2.4. Beyond-accuracy objectives

Kaminskas and Bridge (2016) survey the measures used when recommendation quality is treated as more than predictive accuracy, and give the definitions of novelty and serendipity adopted in Section 4.6: novelty as the self-information of a recommended item, and serendipity as relevance discounted by how expected the item was. Adomavicius and Kwon (2012) show that aggregate diversity can be improved substantially at a small accuracy cost, which is the trade-off this project measures rather than assumes.

2.5. Evaluation methodology

Cremonesi et al. (2010) report that on top- N tasks a non-personalized popularity baseline is far stronger than its reputation, and that models tuned for rating error do not necessarily rank well. Both observations recur in Section 5.4. Meng et al. (2020) show that the choice of data splitting strategy materially changes which model appears best, which motivates the robustness check in Section 5.8. Panda and Ray (2022) review cold-start mitigation and note that the right strategy depends on whether the cold entity is a user or an item and on what side information exists — a distinction Section 5.6 relies on.

3. Data

3.1. MovieLens `ml-latest-small`

All experiments use the MovieLens `ml-latest-small` benchmark (Harper & Konstan, 2015): 100,836 ratings on a half-star scale from 0.5 to 5.0, contributed by 610 users across 9,724 distinct rated films. The full catalogue file lists 9,742 titles; the 18 with no ratings never enter the user–item matrix. That matrix is therefore $610 \times 9,724$ with 98.3% of its cells empty. Every user in the release has rated at least 20 films, so there are no cold *users* here — the cold-start problem measurable in this dataset is entirely item-side.

3.2. TMDB plot overviews

The content model needs text. Plot overviews were fetched from The Movie Database using the `tmdbId` column of the benchmark’s `links.csv`, giving 9,603 of the 9,724 rated films (98.8%). The 121 that remain uncovered are not a collection gap: 113 have no synopsis on TMDB at all and 8 carry no `tmdbId`, so they are not fetchable. Appendix A reports what expanding this coverage did to the content model, which is not what was expected.

3.3. Split protocol

Each user’s ratings are shuffled and split 60/20/20 into train, validation and test. Splitting per user rather than globally guarantees every user appears in all three parts, so no user is unmeasurable at evaluation time. The shuffle is seeded with the author’s student ID, and the test slice is taken before the validation slice is carved out, so a two-way and a three-way split of the same seed select identical test rows. Hyperparameters are selected on validation only; models are then refitted on train+validation and the test set is evaluated exactly once per model.

3.4. Cold items

732 of the 9,724 films receive no rating at all in the combined train+validation portion, and 794 test ratings fall on them. These are the *cold items* referred to throughout. They are 7.5% of the catalogue but only 2.92% of the relevant held-out ratings, and that asymmetry explains much of Section 5.6.

4. Methods

4.1. Matrix factorization (SVD)

The rating matrix is mean-centred, missing entries are filled with zero, and a truncated SVD retains the k largest singular values. Reconstruction adds the centring back and clips to the valid rating range. An item-bias term is added, damped as the sum of an item’s residuals divided by its rating count plus a damping constant, so an item with two ratings is shrunk toward zero bias while one with two hundred is not. Two centring modes are used: subtracting the user mean only, and subtracting the user mean and then the item bias. The validation grid selected $k = 10$ with damping = 5.

The behaviour on cold items follows from the construction and is worth stating because Section 5.6 tests it. An item with no training ratings contributes an all-zero column to the centred matrix, so the reconstruction contributes nothing and the predicted rating is exactly the user’s own mean, for every such item.

4.2. I-AutoRec

The item-based autoencoder (Sedhain et al., 2015) takes each item’s 610-dimensional vector of user ratings as one training example, encodes it through a sigmoid layer of width h , and decodes it linearly. The loss is mean squared error masked to the observed entries, so unobserved cells contribute no gradient. Training uses Adam with weight decay as the regularizer and early stopping on validation error. Two configurations are carried forward because the validation grid selected different ones for the two metric families — $h = 200$ with weight decay 10^{-4} minimised validation RMSE, while $h = 25$ with weight decay 10^{-2} maximised validation NDCG@10 — and reporting one model under both headings would misrepresent it. Appendix B gives the grid.

The same cold-item argument applies here and is sharper. A cold item’s input vector is entirely zero, so the encoder sees the same input for every cold item and the decoder can only emit its own bias. Every cold item therefore receives an identical prediction, for a given user, regardless of which film it is. Non-linearity does not help, because there is no signal for the non-linearity to act on.

4.3. Content-based model

Plot overviews are vectorised with TF-IDF and items compared by cosine similarity. A user profile is the similarity-weighted average of the items they rated, centred on the user’s mean rating so a below-average rating pushes away from an item rather than weakly toward it. The model scores only items that have text, which is 98.8% of the catalogue but never all of it, and unlike the collaborative models it is defined for an item that has never been rated. Genre tokens were tested as an additional feature and rejected on validation (Section 5.7).

Because the model outputs a centred similarity rather than a rating on the 0.5–5 scale, a rating error against it is undefined, not merely unmeasured. The implementation refuses to compute one rather than returning a misleading figure.

4.4. Hybrid composition

Component scores are not comparable in raw form: an SVD reconstruction lives on the rating scale and a content score is a centred cosine similarity. Each component’s scores are therefore standardised per user to zero mean and unit variance before combining, so the blend weight controls the relative influence of the two signals rather than the relative size of their units.

Two of Burke’s (2002) strategies are implemented. The weighted hybrid returns $\alpha z_1 + (1 - \alpha) z_2$ over the standardised component scores. The switching hybrid uses the first component where it has a collaborative signal and the sec-

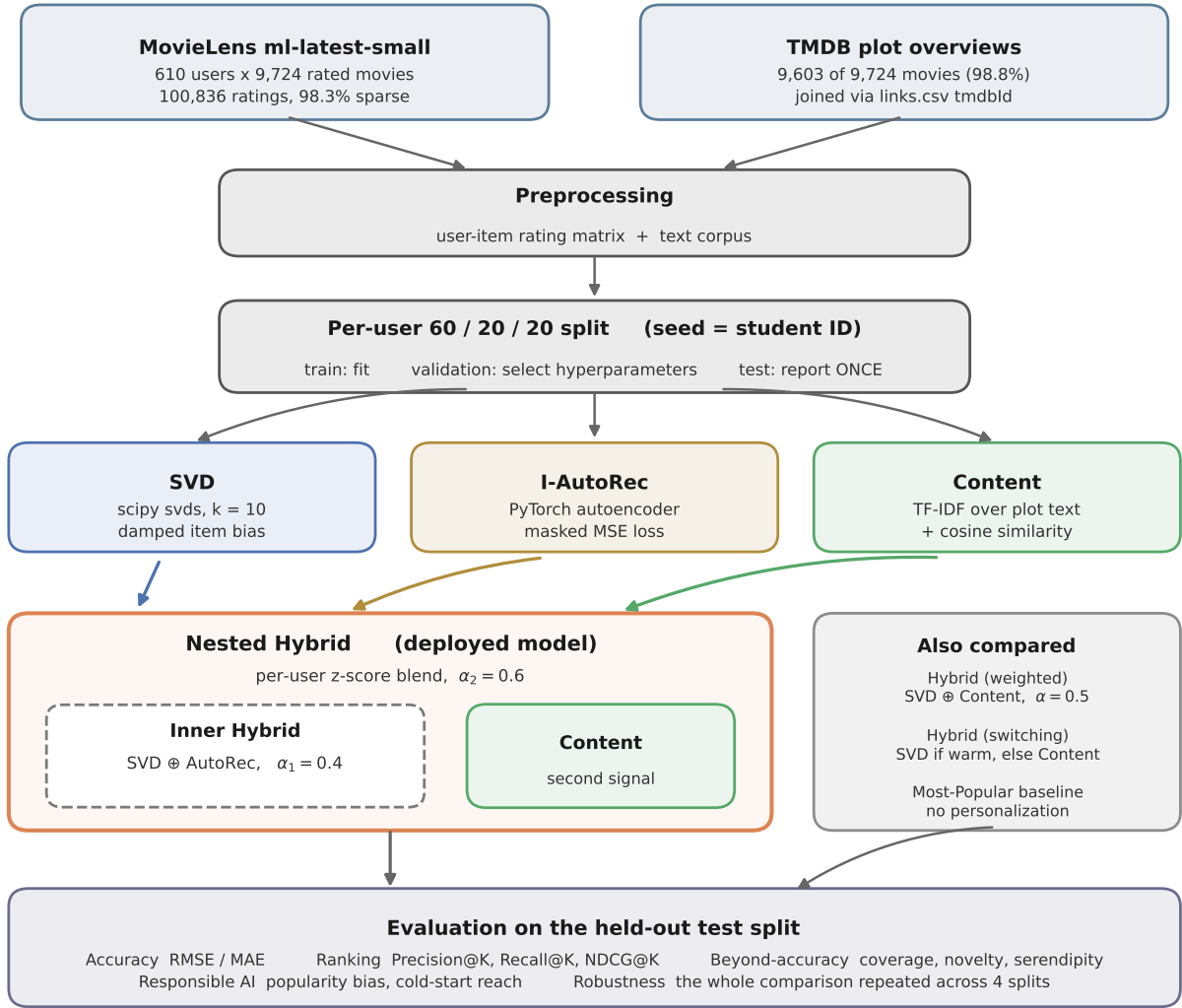


Figure 1. Model architecture. Data, the three-way split, the three base models, the nested composition, the ablation variants and the evaluation battery. The nested hybrid is drawn as a box inside a box because that is what it is: a hybrid whose first component is another hybrid, not a special-cased three-way blend.

and where it does not, which is what makes it the variant with the most cold-start reach in Section 5.6.

The design decision that matters is that a hybrid accepts any component satisfying the same small interface, and a hybrid satisfies that interface itself. A three-model system is therefore a composition of two two-model hybrids rather than a separate implementation:

```
Hybrid( Hybrid(SVD, AutoRec,  $\alpha_1$ ),
        Content,  $\alpha_2$  )
```

The consequence is not only brevity. Because the components stay separate objects, each can still be measured on its own, which is what makes the attribution in Section 5.5 possible. A single model trained end-to-end on all three signals would rank better or worse as one number, with no way

to say which signal produced the difference.

4.5. Baselines

Three baselines are used. The user-mean predictor bounds the rating-error results from below in difficulty. An off-the-shelf scikit-learn `TruncatedSVD` without the item-bias term isolates the contribution of that term. Most-Popular, which ranks every user’s unseen items by training-set rating count, is the ranking baseline; following Cremonesi et al. (2010) it is treated as a serious competitor rather than a formality, and Section 5.4 shows why.

4.6. Evaluation measures

Five families are reported.

- **Rating error.** RMSE and MAE over test ratings the model can score. Only the SVD and AutoRec produce calibrated ratings, so this applies to two of the seven systems.
- **Ranking.** Precision@ K , Recall@ K and NDCG@ K at $K \in \{5, 10, 20\}$, with an item counted relevant if rated ≥ 4.0 in the test split. Precision divides by K rather than by the returned list length, so a model returning a short list is charged for the unfilled slots instead of flattered by them.
- **Catalogue coverage.** The fraction of the catalogue appearing in at least one user’s top-10, reported against the whole catalogue and against each model’s reachable pool.
- **Novelty and serendipity.** Novelty is the mean self-information $-\log_2 p(i)$ of recommended items, with add-one smoothing so the 732 unrated items score the highest finite value rather than infinity. Serendipity is Precision@10 with each hit weighted by $1 - p(i)$; it shares Precision’s denominator and 0–1 scale, so the ratio of the two reads as the share of a model’s accuracy that came from something other than best-sellers.
- **Cold-start reach.** The number of top-10 slots, out of 6,100, occupied by items with no train+validation ratings.

Novelty needs a scale to be read against, and two reference points are reported with it: a uniformly random recommender scores the catalogue average of 7.13, and an item nobody has rated scores the ceiling of 9.26. The single most-rated film scores 1.19.

5. Experiments

5.1. Setup

All randomness is seeded with the student ID, so the split and every reported figure reproduce exactly. Hyperparameters are selected on validation, models are refitted on train+validation, and the test split is evaluated once.

5.2. Hyperparameter selection

The SVD was searched over $k \in \{5, 10, 15, 20, 30, 50\}$ crossed with damping $\in \{1, 5, 10, 25, 50\}$. The content model was tested with and without genre tokens. The blend weight α was swept from 0.0 to 1.0 in steps of 0.1 under two fallback strategies. I-AutoRec was searched over hidden width $\in \{25, 200\}$ crossed with weight decay $\in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$. The nested hybrid’s two alphas were searched jointly. Appendix B reports the grids.

Model	MSE	RMSE	MAE
User-mean baseline	0.8927	0.9448	0.7353
sklearn TruncatedSVD ($k=10$)	—	0.9153	0.7063
SVD ($k=10$, damp=5)	0.7628	0.8734	0.6663
I-AutoRec ($h=200$)	0.7253	0.8517	0.6562

Table 1. Test rating error. The test split is touched once.

Model	@5	@10	@20
SVD	0.1596	0.1502	0.1456
Content	0.0401	0.0359	0.0345
Hybrid (weighted)	0.1731	0.1605	0.1568
Hybrid (switching)	0.1560	0.1474	0.1430
Most-Popular	0.1641	0.1549	0.1562
I-AutoRec	0.1562	0.1432	0.1382
Nested Hybrid	0.1796	0.1643	0.1610

Table 2. NDCG@ K on the test split. Plain SVD does not beat the non-personalized baseline at any of the three cutoffs.

Model	RMSE	NDCG	Cov.	Nov.	Ser.	Cold
SVD	0.8734	0.1502	4.0%	2.34	0.0861	0
Content	—	0.0359	24.9%	6.76	0.0183	481
Hybrid (weighted)	—	0.1605	5.5%	2.49	0.0872	6
Hybrid (switching)	—	0.1474	4.2%	2.40	0.0841	60
Most-Popular	—	0.1549	0.6%	1.65	0.0782	0
I-AutoRec	0.8517	0.1432	0.6%	1.86	0.0736	0
Nested Hybrid	—	0.1643	5.1%	2.42	0.0900	1

Table 3. All seven systems, all five metric families, one split. A dash in the RMSE column is a property of the model, not missing data. Novelty is on a scale where a uniformly random recommender scores 7.13.

5.3. Rating error

Table 1 shows two things. The damped item-bias term is worth 0.0419 RMSE against an otherwise comparable library implementation without it, a larger gain than the difference between several of the models compared later. And the deep model beats the linear one by 0.0217 RMSE. On this evidence alone I-AutoRec is the best model in the project, which is precisely the conclusion the rest of this section undermines.

5.4. Ranking

Plain SVD loses to Most-Popular at all three cutoffs, and I-AutoRec loses to it more heavily. This reproduces the effect Cremonesi et al. (2010) report: a model selected for rating error is not thereby a good ranker, and a popularity baseline is a harder target than it looks. Only the weighted hybrids clear it. Section 5.8 revisits the SVD comparison, because the margin is small enough to be worth testing.

5.5. Beyond accuracy: the central result

Six measures are available — lowest rating error, highest NDCG@10, highest novelty, highest serendipity, widest

coverage, most cold-start reach — and three different models hold them. No ordering of the seven systems is consistent across Table 3, which is this project’s central finding and the reason it does not report a single best model.

The novelty column sharpens what coverage only hints at. I-AutoRec’s coverage of 0.6% is identical to the non-personalized baseline’s, which establishes that it is as narrow; its novelty of 1.86 against that baseline’s 1.65 establishes something stronger, that it is narrow *in the same place*. On a scale where picking uniformly at random scores 7.13, the six accuracy-oriented systems all fall between 1.65 and 2.49. They are not somewhat popularity-biased; they occupy the bottom quarter of the scale. I-AutoRec’s advantage in Table 1 is bought by predicting well on films almost everyone has already rated.

Serendipity tells the same story from the accuracy side. Because it shares Precision@10’s denominator and scale, the ratio of the two is interpretable directly as the share of a model’s hits that were not on best-sellers: 0.678 for Most-Popular, 0.702 for I-AutoRec, rising through 0.739 for SVD and 0.748 for the nested hybrid to 0.851 for the content model. The ordering is the opposite of the ranking ordering in Table 2.

5.6. Cold start is a data problem

The 732 cold items provide a direct test of whether model capacity can compensate for missing signal. For the SVD, the largest absolute difference between the predicted rating on a cold item and the user’s own mean is 0.00e+00 across every such row, and RMSE on those rows equals the user-mean baseline’s exactly, at 0.9805. For I-AutoRec, the standard deviation of a given user’s predictions across all 732 cold items is 0.00e+00, against 0.17 to 0.50 on a sample of 200 warm items for the same users.

A deeper, non-linear model therefore does not solve cold start by being more expressive. Both models are looking at an all-zero input and can only return a constant. This is a mathematical consequence of the architectures rather than a statistical finding, and Section 5.8 confirms it holds unchanged on every split tested.

Content signal is what breaks the tie. The content model places 481 cold items into top-10 lists out of 6,100 slots; the switching hybrid, which hands cold items to the content component instead of averaging them against a flat collaborative score, reaches 60; the weighted hybrid reaches 6; SVD, I-AutoRec and Most-Popular reach none. The cap is the blend weight rather than the text: Appendix A reports that raising text coverage of cold items from 18% to 98% left the weighted hybrid’s cold reach at exactly 6.

5.7. Ablations

Four ablations were run; Appendix B gives the grids.

Ordering	diff	std	Holds
Nested > Hybrid (wtd)	0.0035	0.0008	4/4
Hybrid (wtd) > SVD	0.0103	0.0022	4/4
Nested > Most-Popular	0.0150	0.0076	4/4
Most-Popular > AutoRec	0.0073	0.0050	4/4
SVD > Hybrid (switch)	0.0025	0.0006	4/4
SVD vs. Most-Popular	0.0012	0.0060	sign flips

Table 4. Head-to-head NDCG@10 orderings, differenced within each of four splits. Five of six hold on every split; the sixth does not.

- **Item-bias term.** Removing it costs 0.0419 RMSE (Table 1).
- **Genre tokens.** Adding genre tokens to the TF-IDF features hurt validation ranking quality and was rejected.
- **Blend weight and fallback.** $\alpha = 0.5$ with the neutral fallback was selected on validation NDCG@10.
- **Capacity and regularisation.** Validation RMSE improved monotonically with hidden width up to 200, while validation NDCG@10 was maximised at width 25 with the strongest weight decay tested — the configuration whose validation RMSE, 1.0819, is worse than the user-mean baseline’s. The two metric families select opposite ends of the same grid, which is the clearest single piece of evidence for this paper’s thesis.

5.8. Robustness across splits

Every number above comes from one split. Following Meng et al. (2020), the entire seven-model comparison was repeated on four splits — the reported seed plus three chosen in advance — with hyperparameters held at the values the reported split’s validation set selected. Any single model’s NDCG@10 moves by up to 0.0123 between splits, so two models closer together than that on one split cannot be called separable from that split alone.

Comparing pairwise within each split is the sharper test, because all seven systems share the split and a generous split lifts them together; differencing cancels that shared movement. Table 4 reports the result; every standard deviation in this paper is the sample standard deviation ($n - 1$).

Eight of ten claims tested hold on all four splits. The exception is the SVD comparison from Section 5.4. On the reported split SVD loses to Most-Popular at all three cut-offs, but across four splits the difference averages +0.0012 NDCG@10 and changes sign twice. The defensible claim is therefore not that collaborative filtering is worse than recommending best-sellers but that it cannot be distinguished from it — which motivates the hybrid just as strongly, and is a claim the evidence actually supports.

The hybrid’s own advantage is unaffected: the weighted hybrids beat plain SVD on every split by $+0.0103 \pm 0.0022$.

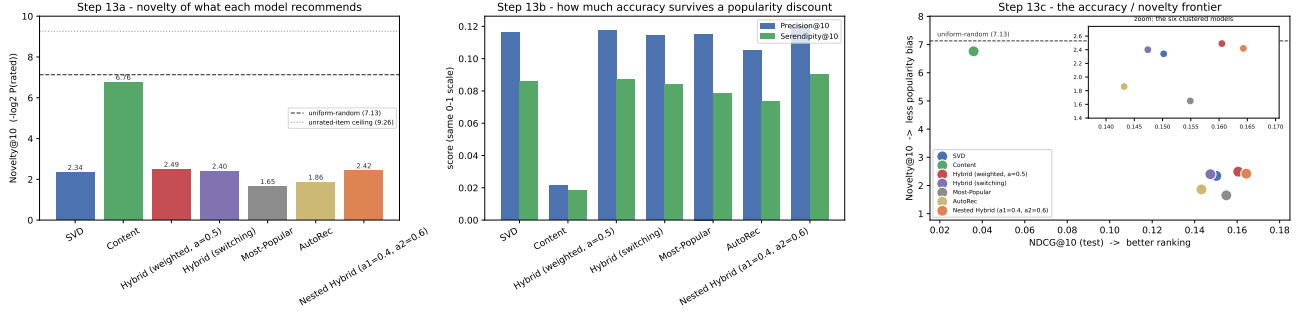


Figure 2. Beyond-accuracy results. Left: novelty of what each model recommends, against the uniform-random line (7.13) and the unrated-item ceiling (9.26). Centre: serendipity beside precision on the same scale. Right: the accuracy/novelty frontier — if any model dominated it would sit alone in the upper right, and none does. The inset zooms the six systems that cluster below the random line.

The beyond-accuracy findings are considerably more stable than the ranking ones — the content model’s coverage is $24.9\% \pm 0.2\%$ and its novelty 6.74 ± 0.03 across all four splits — and the zero-variance cold-item result holds exactly, as the argument in Section 5.6 requires.

5.9. Error analysis

Three failures are worth reporting. First, selection noise is visible at this scale: the nested hybrid’s validation grid was won by a degenerate corner, $\alpha_1 = 0$ and $\alpha_2 = 1$, which reduces the three-model system to I-AutoRec alone. It beat the best genuine three-model blend by 0.0006 NDCG@10 on validation and then lost to it on test by 0.0211, 0.1432 against 0.1643. Differences in the fourth decimal on a validation set of roughly 600 users should not be treated as real, and the reported system uses the interior blend for that reason.

Second, expanding the plot-text corpus improved the content model’s coverage and made its ranking worse. Cold items are 2.92% of the relevant held-out ratings, so promoting one into a top-10 usually costs a slot a popular item would have converted. Offline ranking metrics can credit only items somebody already rated and therefore penalise discovery systematically; this is a limitation of the evaluation, not a regression in the model.

Third, the switching hybrid ranks consistently below plain SVD, by 0.0025 ± 0.0006 across four splits, while reaching 60 cold-start slots against SVD’s zero. That single pair contains the accuracy/coverage trade-off in both directions, and both directions are separable.

6. Conclusion

Seven movie recommenders were built and evaluated on one held-out split under five families of measure. The nested hybrid ranks best (NDCG@10 0.1643) and is the most serendipitous; I-AutoRec has the lowest rating error (0.8517) while recommending from 0.6% of the catalogue, the same fraction as a non-personalized baseline and

from the same end of the popularity distribution; the content model is the most diverse and the worst ranker. Three different models hold the six available titles, so reporting a single best model would have concealed the result rather than summarising it.

Composability mattered as much as any individual model. Because a hybrid can take another hybrid as a component, the three-model system is two lines of composition, and because the components remain separately measurable, the attribution in Section 5.5 is possible at all.

The main limitations are: one dataset and one domain, so conclusions about capacity are tied to this scale; offline evaluation only, which penalises discovery by construction; hyperparameters selected on one validation split, so Section 5.8 measures split variance rather than selection variance; and a serving layer that scores a fixed set of users in advance, so a new user receives the popularity baseline until the next batch run.

Three extensions follow directly. An online or interleaved evaluation would test whether the coverage the content model buys is worth the offline ranking it costs. Repeated splits for hyperparameter *selection*, not only for reporting, would capture selection variance. And feeding content features into the deep model rather than blending them afterwards — a two-tower design, or injecting text similarities into the hidden layer — would address the fact that all 732 cold items have plot text I-AutoRec cannot see. That trade-off runs both ways: late fusion is what keeps each component separately measurable, and it is what allowed this project to show that I-AutoRec collapses onto popularity while the content model does not.

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A. Expanding plot-text coverage

The TMDb corpus originally covered 3,536 of the 9,724 rated films (36.4%), and only 18% of cold items. It was

expanded to 9,603 films (98.8%) and 98% of cold items. The effect on the content model was measured rather than assumed: cold-start reach roughly tripled, from 163 to 481 top-10 slots, and catalogue coverage rose from 17.8% to 24.9%, while ranking quality fell, NDCG@10 going from 0.0469 to 0.0359.

An earlier version of this project predicted that the weighted hybrid’s cold-start reach was capped by text coverage. That prediction was tested and is wrong: coverage of cold items went from 18% to 98% and the weighted hybrid’s reach stayed at exactly 6 slots out of 6,100. The real cap is the blend weight. A cold item has no collaborative signal, so its standardised collaborative score sits at the user’s own mean, and at $\alpha = 0.5$ the content half cannot lift it past warm items scoring well on both halves. The switching hybrid, which does not average against a flat score, is the variant whose reach improved, from 38 to 60 slots.

B. Hyperparameter grids

h	decay	val RMSE	val NDCG@10	epoch
25	10^{-5}	0.9035	0.0382	205
25	10^{-4}	0.9004	0.0436	196
25	10^{-3}	0.8891	0.0830	374
25	10^{-2}	1.0819	0.1196	413
200	10^{-5}	0.8629	0.0656	67
200	10^{-4}	0.8616	0.0721	67
200	10^{-3}	0.8634	0.0853	75
200	10^{-2}	0.9093	0.0942	2831

Table 5. I-AutoRec validation grid. The two metric families select opposite ends of it.

C. Robustness across splits

Model	s1	s42	s12345	reported	std
SVD	0.1529	0.1545	0.1585	0.1502	0.0035
Content	0.0341	0.0351	0.0352	0.0359	0.0007
Hybrid (weighted)	0.1601	0.1664	0.1703	0.1605	0.0049
Hybrid (switching)	0.1504	0.1527	0.1553	0.1474	0.0034
Most-Popular	0.1560	0.1468	0.1536	0.1549	0.0041
I-AutoRec	0.1453	0.1407	0.1530	0.1432	0.0053
Nested Hybrid	0.1638	0.1705	0.1726	0.1643	0.0044

Table 6. NDCG@10 across four splits. Widest swing for any single model: 0.0123.

Coverage and novelty are markedly more stable than ranking. Across the same four splits, coverage is $24.9\% \pm 0.2\%$ for the content model, $5.5\% \pm 0.0\%$ for the weighted hybrid and $0.6\% \pm 0.0\%$ for both I-AutoRec and Most-Popular; novelty is 6.74 ± 0.03 , 2.50 ± 0.02 , 1.84 ± 0.07 and 1.64 ± 0.01 respectively. Rating error is 0.8466 ± 0.0061 for I-AutoRec and 0.8664 ± 0.0077 for the SVD.

Caveat. Hyperparameters were selected on the reported split’s validation set and held fixed, so some of that data lands in another split’s test set. The optimism this introduces is small, because the choices are coarse, but it is real: this measures split variance with the model fixed, not selection variance.

D. Deployment

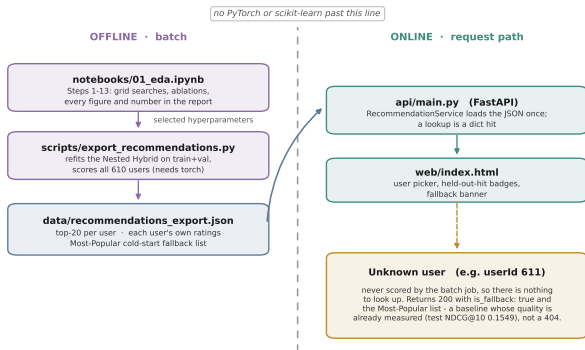


Figure 3. Batch/serving split. The batch job fits and scores; the serving tier reads the resulting table. No training framework is loaded in the request path.

A batch script refits the deployed nested hybrid on train+validation, scores all 610 users and writes a JSON lookup table. The serving process reads that file and never imports PyTorch or scikit-learn, so a recommendation is a dictionary lookup. A user the batch job never scored is answered with the Most-Popular list and an explicit flag rather than an error; because that fallback is the same baseline evaluated in Table 2, its quality is a reported number rather than a guess.

Across all 610 users, 192 recommendations in the top ten are films the user really did rate — in the test split, which the model was never fitted on. 166 of the 610 users see at least one. These are held-out hits rather than leakage, and the interface labels them as such.

E. Declarations

Individual contribution. This project was completed individually. All code, experiments, figures and writing are the author’s own work.